QP CODE

Reg. No:.....
Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP(HONOURS) REGULAR EXAMINATION MARCH 2026
FOURTH SEMESTER
DSE- MG4DSEPHY207- CURRENT ELECTRICITY
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions
Answer any 10 questions
Each question carries 2 marks

1. State the principle of voltage divider circuit. [U] [CO 1]
2. State Kirchhoff's Current Law (KCL). [K] [CO 1]
3. What is meant by a short circuit? Why is a short circuit considered dangerous in household wiring? [U] [CO 1]
4. Why can't superposition be applied directly to power calculations? [An] [CO 2]
5. State the condition for maximum power transfer in a DC network. [K] [CO 2]
6. State Faraday's law of electromagnetic induction. [K] [CO 3]
7. What is a parallel plate capacitor? Write the formula for capacitance of a parallel plate capacitor? [U] [CO 3]
8. Write the expression for resistance of an electrolyte in terms of ρ, length and area? [A] [CO 4]
9. Why do strong electrolytes have high conductivity even at low concentrations? [A] [CO 4]
10. Define Capacitive Reactance (X_C) and explain its dependence on frequency. [A] [CO 5]
11. Show that the power consumed per cycle is zero for a circuit containing inductance only. [An] [CO 5]
12. What is wattless current? Give one example where it occurs. [U] [CO 5]
13. Define the time constant of an LR circuit in which current is decaying. [K] [CO 5]

14. Sketch the nature of the graph between thermo emf and the temperature difference between the hot and cold junction. [A] [CO 6]

(2 × 10 = 20)

Part B

Short Essay Questions
Answer any 6 questions
Each question carries 5 marks

15. Explain the effect of temperature on the resistance of materials. [An] [CO 1]

16. State and explain Thevenin's theorem. How will you Thevenize a given circuit?

[U, A] [CO 2]

17. Two coil A and B are placed near each other and have 200 and 800 turns respectively. A direct current of 2 amperes in coil A produces a flux of 2.5×10^{-4} Wb in A and 1.8×10^{-4} Wb in B. Determine (i) the self-inductance of A (ii) the mutual inductance of A and B (iii) the average emf induced in B when the current in A is interrupted in 0.3 second.

[A] [CO 3]

18. Explain Arrhenius theory and write its main postulates. Mention any limitations of the theory. [A] [CO 4]

19. Discuss the LCR series circuit in detail when connected to an AC supply and bring up the idea of electrical resonance. [An] [CO 5]

20. Define time constant of LR and RC circuits and explain its physical significance.

[An] [CO 5]

21. Deduce an expression for the emf induced in a coil rotating in a uniform magnetic field.

[An] [CO 5]

22. If for a certain thermocouple $E = at + bt^2$ where t °C is the temperature of the hot junction, the cold junction being at 0 °C. $a = 10$ μV/°C, $b = -1/40$ μV/°C, find the neutral temperature and temperature of inversion. [A] [CO 6]

(5 × 6 = 30)